

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:	22/7/26	Duration:	60 minutes

Code of Conduct

I S.PRATEEKSHA-192572137 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S.PRATEEKSHA

EMPLOYEE PERFORMANCE EVALUATIONS AND PRODUCTIVITY ANALYTICS SYSTEM

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. Understand the Problem Statement

- Explain the problem in your own words.
 - Employee performance evaluation is one of the most important functions of Human Resource Management. In many organizations, employee performance is still evaluated using manual records, spreadsheets, or isolated software tools.
 - These traditional methods consume significant time, are prone to human errors, lack transparency, and make it difficult to monitor employee productivity continuously. Managers often struggle to generate accurate reports, identify high-performing employees, and provide timely feedback.
 - The proposed **Employee Performance Evaluations and Productivity Analytics System** is a software-based solution that automates employee performance monitoring, productivity analysis, goal tracking, and report generation
- Identify the objectives of the proposed system
 - Automate employee performance evaluation.
 - Track employee productivity using measurable KPIs.
 - Reduce manual errors and paperwork.
 - Provide real-time dashboards and analytics.
 - Enable managers to make data-driven decisions.
 - Improve employee motivation through continuous feedback.
 - Generate accurate performance reports for promotions and appraisals.
- **Scope of the Problem:**
 - The system covers:
 - Employee profile management
 - Attendance monitoring
 - Task and project tracking
 - Goal and KPI management
 - Performance evaluation
 - Productivity analytics
 - Feedback management

- Performance report generation
- HR administration
- Dashboard visualization

The system is intended for organizations of all sizes and can support multiple departments simultaneously

2. Identify Key Stakeholders and Challenges

○ Identify all the stakeholders involved in the system.

- The Employee Performance Evaluations and Productivity Analytics System involves several stakeholders who contribute to its successful implementation and operation. The primary stakeholders include employees, managers, the Human Resource (HR) department, system administrators, top management, and the IT support team.
- Each stakeholder interacts with the system based on their responsibilities and contributes to achieving the organization's performance management objectives. Employees use the system to update task progress, monitor their performance, and receive feedback, while managers evaluate employee performance and make decisions regarding promotions and development.
- The HR department oversees the appraisal process and maintains employee records, whereas system administrators ensure that the software functions securely and efficiently. Top management utilizes the analytical reports generated by the system to make strategic business decisions, and the IT support team maintains the technical infrastructure and resolves system-related issues

○ Explain the roles and responsibilities of each stakeholder.

- Employees are responsible for updating their work progress, completing assigned tasks within deadlines, maintaining accurate records of their activities, and reviewing the feedback provided by their managers.
- They can also monitor their performance scores and work toward achieving their assigned goals. Managers are responsible for assigning tasks, defining performance objectives, monitoring employee productivity, conducting periodic evaluations, providing constructive feedback, and recommending employees for promotions or training based on their performance.

- The Human Resource department manages employee profiles, establishes performance evaluation criteria, coordinates appraisal cycles, generates organizational performance reports, and ensures that the evaluation process is fair and consistent across all departments. System administrators are responsible for creating user accounts, managing user permissions, maintaining system security, performing database backups, monitoring system performance, and ensuring the availability of the software
 - . Top management uses the reports and productivity analytics generated by the system to evaluate organizational performance, allocate resources effectively, identify high-performing employees, and develop long-term business strategies. The IT support team is responsible for maintaining the software infrastructure, resolving technical issues, performing software updates, ensuring data security, and providing technical assistance whenever required.
- Discuss the major challenges and issues faced by the stakeholders.
- Each stakeholder encounters different challenges in the existing employee performance evaluation process. Employees often experience delays in receiving feedback and may feel that the evaluation process lacks transparency, making it difficult for them to understand how their performance is assessed.
 - Managers face challenges in monitoring the performance of a large number of employees, manually calculating performance scores, and maintaining consistency in evaluations across different departments.
 - The HR department spends considerable time managing paper-based records, consolidating performance data from multiple sources, and preparing reports for management, which increases the possibility of errors and delays. Top management faces difficulties in obtaining real-time insights into organizational productivity, making it challenging to make timely and informed strategic decisions.
 - The IT support team must address issues related to system security, software maintenance, database management, and integration with other enterprise applications.

3. Analyze the Current Approach

- Describe the existing system or current process.
- In many organizations, employee performance evaluation is still carried out using traditional methods such as paper-based appraisal forms, Microsoft Excel spreadsheets, emails, or basic Human Resource Management (HRM) software that lacks advanced analytical capabilities.
 - Managers evaluate employees periodically, usually on a quarterly or annual basis, by reviewing attendance records, completed tasks, and individual achievements. The collected information is manually analyzed before preparing performance reports and appraisal documents.

- Feedback is generally provided only during formal evaluation periods, resulting in limited opportunities for continuous performance improvement. Since data is often stored in different locations or managed by separate departments, retrieving employee information and generating organization-wide performance reports becomes a time-consuming process. As the organization grows, managing employee performance manually becomes increasingly difficult and inefficient.

- **Identify its strengths and limitations.**

Although the current system has several drawbacks, it also provides a few advantages. The strengths and limitations are discussed below.

Strengths

- The system is simple and easy to implement.
- It requires a low initial investment.
- Employees and managers are already familiar with the existing process.
- Small organizations can manage employee records without advanced software.
- Managers can include personal observations during performance evaluations.

Limitations

- Manual data entry increases the possibility of human errors.
- Performance evaluation is time-consuming and labor-intensive.
- Reports are generated slowly due to manual calculations.
- There is no real-time monitoring of employee productivity.
- Employee feedback is often delayed.
- Performance data is scattered across multiple files and departments.
- The system lacks analytical dashboards and visualization tools.
- Decision-making is difficult because accurate and up-to-date information is not readily

- **Analyze the problems or gaps that need to be addressed.**

The current employee performance evaluation process contains several shortcomings that reduce organizational efficiency. A centralized and automated solution is required to overcome these issues.

The major gaps identified in the existing system are as follows:

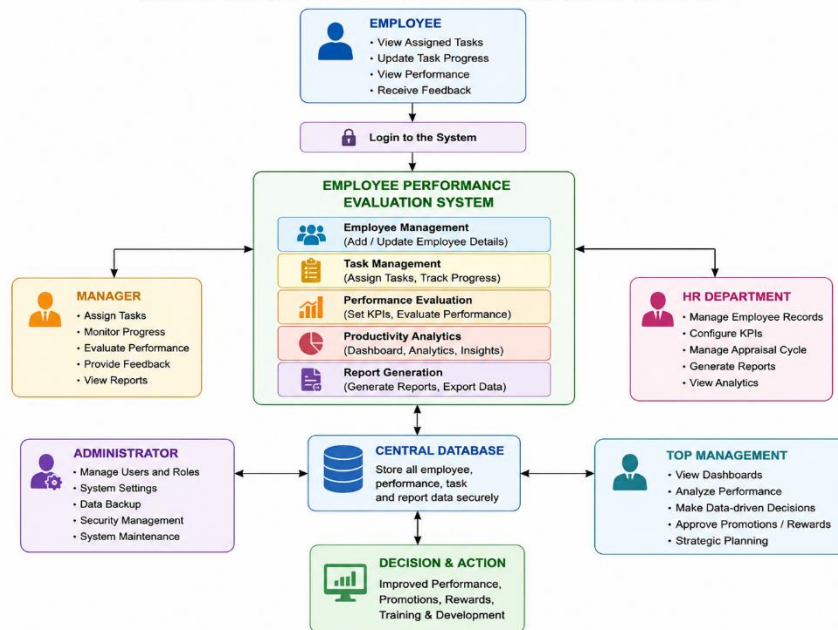
- Absence of a centralized database for employee performance records.
- Manual calculation of performance scores leads to errors and inconsistencies.
- No continuous monitoring of employee productivity.
- Difficulty in tracking Key Performance Indicators (KPIs).
- Delayed feedback affects employee improvement and motivation.
- Limited reporting capabilities make performance analysis difficult.
- Lack of integration with attendance, task management, and HR systems.
- Inability to generate real-time dashboards and productivity analytics.
- Difficulty in identifying high-performing employees using objective data.
- Increased administrative workload for managers and HR personnel.

4. Propose a Suitable Solution Using Software Engineering Principles

- Design a software-based solution for the given problem.

- To overcome the limitations of the existing employee performance evaluation process, a web-based **Employee Performance Evaluations and Productivity Analytics System** is proposed. The system provides a centralized platform for managing employee information, monitoring work progress, evaluating performance, and generating analytical reports.
- It automates the appraisal process by collecting employee performance data, calculating Key Performance Indicators (KPIs), and presenting the results through interactive dashboards. Managers can monitor employee productivity in real time, provide continuous feedback, and make informed decisions regarding promotions, rewards, and training.
- Employees can view their assigned tasks, monitor their performance scores, and receive regular feedback to improve their productivity. The system also enables the HR department to manage appraisal cycles efficiently while providing top management with valuable insights into overall organizational performance.

Figure 1: Employee Performance Evaluation System Architecture



- Describe the major functional modules of the proposed system.

The proposed system is divided into several functional modules, each responsible for performing a specific task. This modular design improves system organization, simplifies maintenance, and allows future enhancements without affecting the entire application.

Module 1: Employee Management

This module maintains employee profiles and organizational information. It allows administrators and HR personnel to register employees, update personal details, assign departments and job roles, and manage employment records.

Module 2: Task and Goal Management

This module enables managers to assign tasks, define performance goals, monitor work progress, and track task completion. Employees can update the status of their assigned tasks, allowing managers to evaluate productivity more effectively.

Module 3: Performance Evaluation

This module evaluates employee performance using predefined Key Performance Indicators (KPIs). It automatically calculates performance scores by considering attendance, task completion, work quality, deadlines, and manager feedback, thereby ensuring consistency and reducing manual errors.

Module 4: Productivity Analytics and Dashboard

This module analyzes employee and departmental performance data to generate dashboards, graphical reports, productivity trends, and comparative analytics. These insights help managers and top management make informed strategic decisions.

Module 5: Feedback and Reporting

This module enables managers to provide regular feedback, conduct performance reviews, generate appraisal reports, and recommend promotions, rewards, or training programs. Reports can be exported in different formats for documentation and future reference.

Module 6: Administration and Security

This module manages user authentication, role-based access control, system configuration, database backup, and audit logs. It ensures that only authorized users can access sensitive employee information while maintaining overall system security.

- Identify the functional and non-functional requirements.

The proposed system must satisfy both functional and non-functional requirements to ensure effective performance and usability.

Functional Requirements

- Employee registration and profile management.
- User authentication and secure login.
- Task assignment and progress tracking.
- Attendance and leave record management.
- KPI definition and performance evaluation.
- Automatic calculation of performance scores.
- Performance feedback management.
- Dashboard and productivity analytics.
- Report generation and export.
- Role-based access control for different users.
- Notification and alert management.

Non-Functional Requirements

- The system should provide fast response times and support multiple users simultaneously.
- It should maintain high reliability and ensure continuous system availability.
- Strong security mechanisms such as password encryption and role-based authorization should protect sensitive employee data.
- The software should be scalable to support organizational growth without affecting performance.
- The system should be maintainable through a modular architecture, making updates and future enhancements .

The proposed Employee Performance Evaluations and Productivity Analytics System is designed by applying key software engineering principles to ensure that it is efficient, reliable, and adaptable to future organizational needs.

Modularity: The system is divided into independent modules such as Employee Management, Task Management, Performance Evaluation, Productivity Analytics, Reporting, and Administration. Each module performs a specific function, making the software easier to develop, test, and maintain.

Scalability: The architecture supports the addition of new employees, departments, performance indicators, and organizational branches without requiring major changes to the existing system. This enables the software to grow along with the organization.

Maintainability: The use of modular programming, proper documentation, and standardized coding practices simplifies debugging, maintenance, and future software upgrades, reducing development costs and improving long-term sustainability.

Reliability: The system incorporates input validation, automated performance calculations, error handling, and regular database backups to ensure consistent operation and accurate results even under heavy workloads.

Usability: A user-friendly interface with intuitive navigation, responsive dashboards, and easy access to reports enables employees, managers, HR personnel, and administrators to use the system efficiently with minimal training.

Security: Sensitive employee information is protected through secure authentication, encrypted passwords, role-based access control, audit logs, and regular database backups. These security measures ensure confidentiality, integrity, and availability of organizational data while preventing unauthorized access.

5. Justify the Chosen Methodology/Framework

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
 - For the proposed **Employee Performance Evaluations and Productivity Analytics System**, the **Agile Software Development Methodology using the Scrum framework** is the most suitable approach. Agile is an iterative and flexible software development methodology that focuses on delivering the software in small, manageable increments called sprints.
 - Instead of developing the entire system at once, Agile allows the development team to complete one functional module at a time, such as Employee Management, Task Management, Performance Evaluation, Analytics Dashboard, and Reporting.
 - After each sprint, feedback is collected from stakeholders, and improvements are incorporated into the next iteration. This approach ensures that the software continuously evolves according to organizational requirements and changing business needs.
- Justify why the selected methodology/framework is suitable for the proposed system.
 - The Agile Scrum methodology is well suited for this project because employee performance evaluation systems often require frequent updates due to changes in organizational policies, Key Performance Indicators (KPIs), appraisal methods, and user requirements.
 - Agile provides the flexibility to accommodate these changes without significantly affecting the overall development process. Since managers, HR personnel, and top management may suggest new features or modifications during development, Agile enables the development team to implement these changes quickly through successive sprints.
 - Furthermore, Agile promotes continuous communication and collaboration among developers, HR staff, managers, and other stakeholders. Regular sprint reviews ensure that the developed modules meet user expectations before proceeding to the next phase. Continuous testing during each sprint also helps identify and resolve defects early, reducing development risks and improving software quality. As a result, the final system becomes more reliable, user-friendly, and aligned with organizational objectives.
- Explain how it supports successful project development and maintenance.
 - Agile supports the successful development and long-term maintenance of the proposed system by emphasizing iterative development, continuous improvement, and customer collaboration. The project is divided into smaller development cycles, allowing each module to be designed, developed, tested, and deployed independently.

- This reduces project complexity and enables the development team to deliver functional software within shorter timeframes. The methodology also encourages continuous stakeholder involvement, ensuring that feedback from employees, managers, and HR personnel is incorporated throughout the development process
- . Regular testing during every sprint improves software quality by detecting defects at an early stage, thereby reducing maintenance costs after deployment. The modular nature of Agile development also simplifies future upgrades, allowing new features such as AI-based performance prediction, advanced analytics, or mobile application support to be integrated without disrupting existing functionality.

Key Advantages of Agile (Scrum) for This Project

- Supports changing business and organizational requirements.
- Delivers functional modules through short development sprints.
- Encourages continuous stakeholder feedback and collaboration.
- Improves software quality through regular testing and reviews.
- Detects and resolves defects early in the development process.
- Reduces project risks and development costs.
- Enables faster deployment of usable software.
- Simplifies future maintenance, upgrades, and feature enhancements.
- Enhances customer satisfaction by delivering software that meets user expectations.
- Supports scalability as the organization and system requirements grow.

Overall, the Agile Scrum methodology provides the flexibility, efficiency, and continuous improvement required for developing an Employee Performance Evaluations and Productivity Analytics System. Its iterative development process, strong stakeholder involvement, and focus on delivering high-quality software make it the most appropriate Software Development Life Cycle (SDLC) model for ensuring the successful implementation, maintenance, and future expansion of the proposed system.

6. Discuss Expected Outcomes and Limitations

○ Describe the expected benefits of the proposed solution.

- The implementation of the Employee Performance Evaluations and Productivity Analytics System is expected to transform the way organizations manage employee performance and productivity.
- By automating the evaluation process, the system reduces the dependency on manual calculations and paper-based records, thereby minimizing errors and saving valuable time for managers and HR personnel. Employees receive timely feedback on their performance, which helps them identify their strengths and areas that require improvement.
- The centralized database ensures that employee information is stored securely and can be accessed whenever required. The analytical dashboards and reports provide

valuable insights into employee productivity, enabling management to make informed decisions regarding promotions, rewards, training, and workforce planning. Overall, the proposed solution enhances transparency, improves employee engagement, increases organizational efficiency, and supports better decision-making through accurate and reliable performance data.

The expected benefits of the proposed system include:

- Improved accuracy in employee performance evaluation.
 - Automated performance monitoring and report generation.
 - Reduced manual workload for managers and HR personnel.
 - Real-time productivity tracking through dashboards.
 - Faster and more transparent appraisal process.
 - Better decision-making based on data analytics.
 - Increased employee motivation through continuous feedback.
 - Enhanced organizational productivity and operational efficiency.
 - Secure storage and management of employee information.
 - Improved communication between employees, managers, and HR
- Explain how the solution addresses the identified challenges.
- The proposed software solution effectively addresses the challenges identified in the existing system by introducing automation, centralized data management, and real-time analytics. Instead of relying on manual calculations and scattered records, the system stores all employee-related information in a centralized database, making data management more efficient and reliable.
 - Automated performance calculations based on predefined Key Performance Indicators (KPIs) eliminate inconsistencies and reduce human errors. Managers can monitor employee performance continuously, provide timely feedback, and identify productivity issues before they become significant problems.
 - The HR department benefits from automated report generation and simplified appraisal management, while top management gains access to interactive dashboards that support strategic decision-making. By improving transparency, reducing administrative effort, and ensuring fair evaluations, the proposed system successfully overcomes the limitations of the traditional performance evaluation process.

- Discuss potential risks, implementation challenges, and limitations.
 - Although the proposed system offers numerous advantages, certain risks and implementation challenges may arise during its deployment. Employees and managers may initially resist adopting the new system because they are accustomed to existing manual processes.
 - Migrating employee records from legacy systems to the new platform may require careful planning to prevent data loss or inconsistencies. Proper training must also be provided to ensure that all users can effectively operate the system.
 - Since the application stores confidential employee information, strong security measures are essential to protect the data from cyber threats and unauthorized access. Additionally, regular software maintenance and system updates are necessary to ensure smooth operation and accommodate changing organizational requirements.

Some limitations of the proposed system are:

- High initial development and implementation costs.
- Requirement for user training and change management.
- Dependence on reliable internet and network infrastructure.
- Need for continuous software maintenance and technical support.
- Data accuracy depends on the correctness of user input.
- Integration with existing organizational systems may require additional resources

- Suggest possible future enhancements.
 - The proposed system can be further enhanced by integrating advanced technologies that improve performance evaluation and decision-making. Artificial Intelligence and Machine Learning can be incorporated to predict employee performance trends, recommend personalized training programs, and identify employees who are at risk of low productivity.
 - A mobile application can provide employees and managers with convenient access to the system from any location. Integration with payroll systems, biometric attendance devices, and Human Resource Management Systems (HRMS) can create a comprehensive workforce management platform. Cloud deployment can improve scalability and accessibility, while advanced business intelligence tools can provide deeper insights into organizational performance and employee development.

Possible future enhancements include:

- Artificial Intelligence (AI)-based performance prediction.
- Machine Learning for advanced productivity analytics.
- Mobile application for remote access.
- Integration with payroll and HR management systems.
- Biometric attendance system integration.
- Cloud-based deployment for improved scalability.
- Automated notifications and reminders.
- Chatbot support for employee assistance.
- Predictive analytics for employee retention and career growth.
- Advanced dashboards with business intelligence and data visualization.

Conclusion

- The proposed **Employee Performance Evaluations and Productivity Analytics System** offers a comprehensive software solution for automating employee performance management and productivity analysis.
- It replaces manual processes with an efficient, secure, and centralized platform that improves accuracy, transparency, and organizational efficiency. By applying software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security, the system provides a robust foundation for long-term organizational growth.
- The adoption of the Agile (Scrum) methodology further ensures flexibility, continuous improvement, and easy maintenance, making the system capable of adapting to changing business requirements. With future enhancements such as Artificial Intelligence, Machine Learning, and cloud integration, the proposed solution can evolve into a complete intelligent workforce management system that supports both employee development and organizational success.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.

Current System Analysis	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.